Rotation certificates and affine period families

A shared supplement to the complete marked-kernel theory

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The complete common paper [3] remains an acquired premise, including AL.1–AL.2, AA.ParabolicPeriods, UA.K, UA.SK.Helly and UA.Geometry. This supplement adds a finite rotation certificate and moves the complete previously accepted cubic, square and finite presentation proofs here so that both [1] and [2] use a common proof. No application is an uncharged premise of another. The exact migration map preserves every moved paragraph and its whole program and data dependencies. The constructions and prior principal results remain attributed to their source authors. The finite certificates are required computations, whereas sample calculations never establish an unbounded parameter assertion.

1 Exact rotation presentations and coset certificates

The following certificates isolate the proof shared by the finite regular and chiral rows. It does not identify a finite image with a presented group, and it does not infer flag transitivity from subgroup intersections alone. Every source-specific table row remains a separate certificate obligation.

Proposition 1 (Exact regular completion)**.** *Let $r \geq 3$, let $P = \langle a_1, \dots, a_{r-1} \rangle$ be a finite group with its specified marking, and put $a_0 = 1$. Suppose that simultaneous inversion of the a_i extends to an automorphism σ of P. Form $H = P \rtimes \langle t \rangle$, with $t^2 = 1$ and $ta_it = a_i^{-1}$, and put $\rho_i = ta_i$ for $0 \leq i < r$. Suppose that these involutions have the full marked subgroup intersection property and that their maximal parabolics $H_i = \langle \rho_j : j \neq i \rangle$ have typed coset Helly. Then the standard H-coset geometry is a regular hypertope with full type-preserving group H, of order $2|P|$. It is isomorphic to the source rotation coset system with parabolics*

$$P_i = \langle a_j^{-1} a_k : j, k \neq i \rangle.$$

Thus P has exactly two orbits on its chambers, and H is its regular completion. For $i \geq 1$ the displayed P_i equals $\langle a_j : j \neq i \rangle$, and $P_0 = \langle a_1^{-1} a_j : j \geq 2 \rangle$.

Proof. The ρ_i are involutions and generate H, since $\rho_0 = t$ and $\rho_0 \rho_i = a_i$. The sign homomorphism $H \to C_2$ has kernel P. Its restriction to H_i is onto, and its kernel is generated by the even words $\rho_j \rho_k = a_j^{-1} a_k$. Hence $H_i \cap P = P_i$ and $[H_i : P_i] = 2$. Every right coset of H_i has an even representative, so $P_i g \mapsto H_i g$ is a bijection on elements of each type. It also preserves incidence. One implication is immediate. Conversely, if two H-cosets have a common odd element h, choose a label k different from their two types. Then $\rho_k h$ is an even common element. This uses $r \geq 3$ and proves the claimed incidence isomorphism without assuming equality of arbitrary products or quotient parabolics.

Full marked IP and typed coset Helly give the regular hypertope by the common chamber construction: Helly gives flag completion and flag transitivity; IP gives trivial chamber stabilizer and two elements in every panel; the generating property in each marked subgroup gives residue connectivity. A type-preserving automorphism fixing a chamber fixes each adjacent chamber by thinness, and then all chambers by gallery connectivity. Consequently the displayed chamber-regular H is the full type-preserving automorphism group. Its index-two subgroup P has exactly

two chamber orbits. The formulas for P_i follow by including $a_0 = 1$ when $i \neq 0$, and by taking differences from a_1 when $i = 0$. □

An exact presentation certificate must first identify P itself. In the finite rows, a completed coset enumeration or an exact fp-group isomorphism supplies an upper bound; an actual permutation image supplies the matching lower bound. A numerical image of the expected order by itself supplies only the lower bound. The regularity test then checks all marked subgroup intersections and the separate normalized coset-Helly identities in H. Neither a group order nor a source structural abbreviation replaces these checks.

For a chiral row the last proposition is not applicable. We must instead certify the rotation coset geometry itself, including thinness, residue connectivity, two chamber orbits with adjacent chambers in opposite orbits, and the absence of simultaneous inversion. Its full automorphism order is then $|P|$, whereas a regular row has full order $2|P|$. The source tables and the literal rotation presentations must retain this distinction.

A bounded certificate for chiral rows

Here all cosets are right cosets. For a finite group P with its three rotation generators, use the four subgroups P_i above and write $K_J = \bigcap_{j\in J} P_j$. For $i \notin J$, choose $j_0 \in J$ and a right transversal T of $P_i \cap P_{j_0}$ in P_{j_0}. Set

$$c(i, J) = \#\{g \in T : g \in P_i P_j \text{ for every } j \in J\}.$$

This counts exactly the type-i elements incident with the base flag of types J, provided that base flag has been fixed; the representatives lie in one finite proper parabolic, not in the entire ambient group.

Proposition 2 (Two chamber orbits, with independent connectivity checks)**.** *Suppose the marked pair-difference subgroups have the full C^+ intersection property. Suppose additionally that*

1. *$c(i, J) = [K_J : K_J \cap P_i]$ for every $|J| = 2$, and $c(i, J) = 2$ for every $|J| = 3$;*
2. *P is generated by the four P_i, and each P_f is generated by its subgroups $P_f \cap P_j$, $j \neq f$;*
3. *for each pair of types F, the graph with two vertex sets indexed by K_F, one for each remaining type i, j, and edges*

$$h \sim k \quad \Longleftrightarrow \quad hk^{-1} \in P_i P_j \qquad (h, k \in K_F)$$

is connected.

Then the rotation coset system is a hypertope. The group P acts freely with two chamber orbits, and adjacent chambers lie in opposite orbits. It is chiral with full type-preserving group P if simultaneous inversion of the three marked generators does not extend to an automorphism of P.

Proof. Pair flags are transitive in any coset system. The $|J| = 2$ equalities say that all eligible cosets are supplied by K_J, so all three-type flags are transitive as well. The C^+ intersection identities give trivial three-type stabilizers: they are the pair-difference group on the one remaining label. The $|J| = 3$ equalities therefore give two completions of each panel, with trivial chamber stabilizer. Every shorter flag extends successively to a chamber, so this is a thin geometry with exactly two P-orbits on chambers. Two adjacent chambers cannot lie in the same orbit, since an element mapping one to the other would fix their panel and hence would be the identity.

The incidence graph of a coset system is connected when its defining subgroups generate its ambient group. Applied first to P, and then to P_f with its subgroups $P_f \cap P_j$, the second condition proves connectivity of the whole geometry and of every rank-three residue. The latter residue identification uses transitivity on three-type flags, so incidence of two elements through the fixed type-f element is represented by an intersection inside P_f. For a rank-two residue, the same transitivity represents each of its two vertex types by K_F; three-type stabilizers are trivial, so these labels are unique. Its actual incidence is exactly the graph in the third condition. This proves residual connectivity independently of the intersection checks.

Thinness and chamber connectivity make every chamber-fixing automorphism trivial. Thus the full type-preserving automorphism group has order at most $2|P|$. If it properly contains P, the latter has index two and is normal. Composing an extra automorphism with an element of P gives an involution t sending the base chamber C to its type-zero neighbor C^0: its square fixes C. For the source coset conventions,

$$C^0 = (P_0 a_1^{-1}, P_1, P_2, P_3).$$

Indeed $P_0 a_i^{-1} = P_0 a_1^{-1}$ for every $i > 0$; these cosets meet all three other base elements, and C^+ intersection excludes $a_1 \in P_0$. Right multiplication gives $C^0 a_i = C^i$, whence $C a_i^{-1} = (C^0)^i$. As type-preserving automorphisms commute with adjacency, $t a_i t$ and a_i^{-1} have the same effect on C and are equal. Thus an extra automorphism would induce simultaneous inversion on P. Its failure proves that P is the full group and the geometry is chiral. □

Every test above is finite and exact. The rank-two connectivity tests use only two copies of an intersection of proper parabolics. A bounded count is not extrapolated to untested presentations. For each literal finite presentation, simultaneous inversion can be decided by evaluating its full relator list on the inverse marked images: if all relations hold, the assignment defines an endomorphism whose square fixes every generator, hence an automorphism; a failed relation excludes it.

2 Literal local periods and actual marked quotients

A period appearing among global relators need not be the least period of the corresponding parabolic. This distinction is checked explicitly. For a finite local group generated by u, v, take one of the three standard rotation presentations

$$\begin{aligned} T_{63}(a,b) &= \langle u, v \mid u^6, v^3, (u^{-1}v)^2, (v^{-1}u^{-2})^a (vu^2)^b \rangle, \\ T_{36}(a,b) &= \langle u, v \mid u^3, v^6, (u^{-1}v)^2, (u^{-1}v^{-2})^a (uv^2)^b \rangle, \\ T_{333}(a,b) &= \langle u, v \mid u^3, v^3, (u^{-1}v)^3, (uv^{-1}u)^a (uv)^b \rangle. \end{aligned}$$

These retain the exact source parameter conventions. Their standard orders are $6(a^2+ab+b^2)$, $6(a^2+ab+b^2)$ and $3(a^2+ab+b^2)$. The usual faithful Euclidean triangle realization is retained here; for each finite row the program also computes the abstract local order independently and tests the exact marked epimorphism into the parabolic. Consequently no local isomorphism is inferred from an unmarked order match.

Let U, V denote the two translation words in the last relator of the chosen presentation. The program tests that their actual images commute, generate a normal subgroup L, and have index six (or three) in the local group. It determines $m = |U|$ and the least $b > 0$ for which $U^a V^b = 1$ with $0 \le a < m$. Then

$$\ker\big(\mathbb{Z}^2 \longrightarrow L,\ (i,j) \longmapsto U^i V^j\big) = \mathbb{Z}(m,0) + \mathbb{Z}(a,b).$$

Indeed reduce the second coordinate modulo b using (a,b) and then the first modulo m. Minimality leaves exactly mb different images; the program independently verifies $mb = |L|$. This

proves the entire lattice kernel, not merely two relations. The order of the complete abstract local presentation after adjoining U^m and U^aV^b is then computed and compared to the actual parabolic order. The generator map is onto and the equal orders prove an exact marked isomorphism.

For spherical rank-three residues the same method uses the complete triangle presentation $\langle u, v \mid u^p, v^q, (u^{-1}v)^r\rangle$, with $1/p + 1/q + 1/r > 1$, and an explicit onto generator map. Its exact order is computed. Every local result therefore specifies its marking and not just a name or a number. The full coset tests in the preceding section independently ensure that these are the actual residues of the global geometry.

When a printed word does not impose its claimed torus period, it is kept literally in the global presentation. The certificate computes the actual local lattice from the three triangle relations and the actual finite parabolic. It never inserts an intended torus word silently. Such a case is recorded separately from a proper collapse of an otherwise valid local period presentation.

3 Rotation presentations and the three shared marking adapters

Let W have involutory Coxeter generators $r_0, \dots, r_3$ and pair orders m_{ij}. Its sign map sends every generator to the nonidentity element of C_2. Reidemeister–Schreier rewriting with transversal $1, r_0$ gives the exact presentation of its even subgroup on $a_i = r_0 r_i$:

$$a_i^{m_{0i}} = 1, \qquad (a_i^{-1}a_j)^{m_{ij}} = 1.$$

The other Schreier generators are their inverses; rewriting both cosets of each Coxeter relator gives conjugates or inverses of these same relators. This proves a presentation identity, not a finite-image argument.

Suppose additional local rotation relators impose regular torus periods $(q, 0)$ or (q, q). Their normal closure inside each local triangle rotation group is invariant under its reflections: in the retained Euclidean lattice realization these are exactly the reflection-invariant period lattices. Their normal closure N in W^+ is then invariant under every global odd generator as well. For a local period and an odd local reflection r_j, write $r_0 = (r_0 r_j) r_j$; the first factor is even and the second preserves the local period subgroup. Thus $N \lhd W$, and

$$(W/N)^+ = W^+/N.$$

Consequently the displayed source rotation presentation, with these local period words, is exactly the even subgroup of the corresponding full reflection presentation. No additional, unverified reflected relation is being discarded.

This supplies the following explicit marking adapters for the shared families. All letters on the right are the involutions in the complete proof sections below; their specified order is preserved.

Family	x	y	z
Cubic (a, b, d, c)	ba	bc	bd
Square (a, b, c, d)	ad	ab	ac
Finite (a, b, c, d)	ca	cd	cb

For the cubic family the two period words become $c(bd)^2b$ and $a(bd)^2b$, with exponents two and s. For the square family, its four toroidal relations become, respectively,

$$(b(ad)^2a)^2, \quad (c(da)^2)^2, \quad (d(cb)^2)^{2s}, \quad (a(bc)^2b)^2.$$

For the finite family they become

$$(d(ca)^2c)^2, \quad (b(ac)^2a)^2, \quad (a(bd)^2)^2, \quad (c(db)^2d)^{3t} \quad (t = 1, 2).$$

These identities follow by cancellation of adjacent involutions, using the indicated commuting pair relations. In the square (s, s) relation, the two translation words commute, so their product raised to s equals $(d(cb)^2)^{2s}$. The parameter t here is exactly the shared finite theorem's parameter. Thus both universal presentations and the ordered local markings, not just their orders, transfer from the common proofs. Their subgroup intersections and flag geometry are unchanged under these explicit marker permutations.

4 The cubic family and its period lattice

Write $a = \rho_0, b = \rho_1, d = \rho_2, c = \rho_3$. Let E be generated by

$$a(x) = (x_2, x_1, x_3), \qquad b(x) = (x_1, x_3, x_2),$$
$$c(x) = (-x_2, -x_1, x_3), \qquad d(x) = (-x_2, -x_1, 1 - x_3).$$

Products act from right to left. Let Q be the signed permutation group in three coordinates, $\epsilon(q) \in \{0, 1\}$ the parity of its number of sign changes, and put $D = \{v \in \mathbb{Z}^3 : \sum v_i \equiv 0 \pmod 2\}$, $A = \{v \in \mathbb{Z}^3 : \sum v_i = 0\}$.

Lemma 1. *The group E is precisely $\{(q, v) \in Q \ltimes \mathbb{Z}^3 : \sum v_i \equiv \epsilon(q) \pmod 2\}$, with translation kernel D. Its marked presentation is*

$$\langle a, b, d, c \mid a^2, b^2, d^2, c^2, (ab)^3, (ad)^2, (ac)^2, (bd)^6, (bc)^3, (dc)^2, (c(bd)^2b)^2 \rangle.$$

Adjoining $(a(bd)^2b)^s = 1$ gives exactly $E_s = E/sD$, of order $48s^3$.

Proof. The linear maps generate all 48 signed permutations. Each affine map satisfies the parity condition, which multiplication preserves. Direct multiplication gives $u = a(bd)^2b = (1, (1, -1, 0))$; its Q-conjugates are the twelve roots $\pm e_i \pm e_j$, generating D. Hence the translation kernel is D. Surjectivity to Q proves the description of E. All eleven presentation relators hold.

Identification of the abstract presentation is separate. The complete program `cubic/kernel_certificate.g` forms its epimorphism onto the specified 48-element group and computes the actual kernel presentation by Reidemeister–Schreier rewriting and exact presentation simplification. It has three generators f_1, f_2, f_3 and exactly

$$[f_2, f_3], \qquad f_3 f_1^{-1} f_3^{-1} f_1, \qquad [f_1, f_2]$$

as relators. The full preimage words are emitted and independently evaluated by `cubic/certificate.py`. Their translations are $(0, 1, 1), (1, 0, -1), (-1, 0, -1)$, of determinant of absolute value 2. Thus this is the whole free abelian kernel, not its abelianization, and its image is D. AL.2 identifies the presented group with E. The exact GAP rewriting and simplification are declared software trust. Conjugating u^s generates exactly the already normal lattice sD, proving the quotient statement and its order. □

5 All parabolics and uniform intersections

Write $E_J = \langle \rho_i : i \in J \rangle$. A mask m denotes the set of positions of its nonzero binary digits. The only infinite proper parabolic is

$$E_{012} = \{(q, v) : q \in S_3 \times \{\pm I\},\ \sum v_i = \epsilon(q)\}, \tag{1}$$

where ϵ is zero on permutations and one on their negatives. Indeed each of a, b, d satisfies this formula, their linear image has order 12, and the conjugates of u by a, b generate A. This proves equality of every fiber. In particular $A \cap sD = sA$.

All other proper parabolics inject into their linear images. Complete closure of the actual affine transformations gives

m	0	1	2	3	4	5	6	8	9	10	11	12	13	14
$\lvert E_J\rvert$	1	2	2	6	2	4	12	2	4	6	24	4	8	48.

The maximal finite groups fix respectively $p = (-1/2, 1/2, 1/2)$ for E_{123}, $p_1 = (0, 0, 1/2)$ for E_{023}, and 0 for E_{013}. The first has linear image Q, and the last is the even-sign subgroup S_4.

Proposition 3. *For every $s \geq 3$ all 256 marked subgroup intersection identities hold in E_s. Its maximal parabolic orders, in omitted-label order $0, 1, 2, 3$, are $48, 8, 24, 12s^2$.*

Proof. For the fourteen finite supports the certificate checks all 196 exact intersections $E_J \cap E_T = E_{J\cap T}$. For any two elements of these groups with equal linear part, each coordinate of the translation difference has absolute value at most 1. Such a difference belongs to sD only if zero for $s \geq 3$, so these meets survive the quotient.

For every finite support J the certificate also checks $Q_J \cap Q_{012} = Q_{J\cap 012}$, and that each actual affine element of E_J with linear part in Q_{012} satisfies (1). These are exactly the elements of $E_{J\cap 012}$, by injectivity of the finite linear images. Any element of the image of E_J also lying in the image of E_{012} has its linear part in this intersection; it therefore already comes from $E_{J\cap 012}$. Identical supports and the full group give the remaining immediate meets. The orders follow from the table and $[A : sA] = s^2$. □

The toroidal maximal sections are exactly $\{3, 6\}_{(2,0)}$ on (c, b, d) and $\{3, 6\}_{(s,0)}$ on (a, b, d), not just maps with their diagrams. The standard Euclidean reflection group $[3, 6]$ has translation lattice A, generated by the conjugates of $u = a(bd)^2b$; the imposed lattice is exactly sA. This is the usual ordered parameter $(s, 0)$. For the first section the corresponding word with a replaced by c has order two. It is a quotient of $[3, 6]_{(2,0)}$, whose lattice calculation gives order $12 \cdot 2^2 = 48$. Our subgroup has order 48, so the marked epimorphism is an isomorphism. The Euclidean reflection presentation and its torus parameter convention are explicitly retained classical facts. The other maximal sections are C_2^3 and the tetrahedral string group S_4.

6 A faithful action and the literal CPR diagrams

Let

$$w_0 = (1, 1, 1),\ w_1 = (1, -1, -1),\ w_2 = (-1, 1, -1),\ w_3 = (-1, -1, 1).$$

The $4s$ labels (i, k), with $k \in \mathbb{Z}/s\mathbb{Z}$, represent plane families $w_i \cdot x = 2k + 1/2$ modulo period $2s$ in the constant. If $qw_i = \eta w_j$, then $\eta = (-1)^{\epsilon(q)}$ is independent of i. An element (q, v) acts by

$$(i, k) \longmapsto \left(j, \eta k + \frac{\eta - 1 + 2w_j \cdot v}{4}\right). \tag{2}$$

The parity condition makes the fraction integral; translation by sD is trivial. The generators a, b, c transpose family pairs $(1, 2), (2, 3), (0, 3)$ respectively, preserving k. The map d swaps families $1, 2$ with $k \mapsto -k - 1$, and preserves $0, 3$ with $k \mapsto -k$.

Lemma 2. *For $s \geq 3$ this is a faithful action of E_s, with exactly the two parity-dependent CPR diagrams on [1, p. 6].*

Proof. An element fixing all four family labels has $q = I$ or $-I$: these are the kernel of the signed-permutation action on the four diagonal lines. For $q = -I$, fixing both levels $0, 1$ forces $s \mid 2$, impossible. For $q = I$ triviality says $w_i \cdot v \in 2s\mathbb{Z}$ for every i. The vectors $(w_0 + w_i)/2$, $i = 1, 2, 3$, are the coordinate vectors, so $v \in s\mathbb{Z}^3$. The condition on w_0 then gives $v/s \in D$. Thus the kernel in E is exactly sD.

The following explicit labeling was read against the source graphics. Begin with $(0,0)$–$(3,0)$, an edge of label 3. Append $s-1$ alternating four-vertex blocks, each joined to the preceding last vertex by label 1:

$$\begin{aligned} \text{diamond } h: &\quad (2,h),(1,h),(1,-h-1),(2,-h-1),\\ \text{square } h: &\quad (3,-h-1),(0,-h-1),(0,h+1),(3,h+1). \end{aligned}$$

In each diamond label 0 joins first–second and third–fourth, and label 2 joins first–third and second–fourth. In each square label 3 joins first–second and third–fourth; label 2 joins first–fourth and second–third. For $s=2t$, finish with label 1 to $(3,t)$ then label 3 to $(0,t)$. For $s=2t+1$, finish with label 1 to $(2,t)$ then parallel labels $0,2$ to $(1,t)$. All levels are modulo s. The alternating levels $0,-1,1,-2,2,\ldots$ show every label occurs once. The explicit generator formulas give precisely the source's two drawings. This is uniform for both parities; the separately run examples are checks of the transcription, not a proof of the family. IP makes these faithful permutation graphs CPR graphs. □

The range matters: at $s=2$ the nontrivial element $(-I,(1,1,-1))$ fixes all eight labels. No faithful degree-eight claim follows.

7 Flag transitivity

Let $P_i=(E_s)_{I\setminus\{i\}}$. For three subgroups X,Y,Z, Tits' elementary coset criterion says that

$$(X\cap Y)(X\cap Z)=X\cap YZ \tag{3}$$

is equivalent to the three corresponding coset types having the Helly property. First, each P_i has this local property on its three maximal parabolics. The certificate checks (3) on the finite pair subgroups. For supports $7,11,13,14$ the product sizes are $12,12,8,36$ and the intersection size is 4 in each case. The equal-linear-part translation-difference bounds are $1,0,0,0$, so all four identities survive modulo sD for every $s\geq 3$.

For triples containing $P_1=\langle a,d,c\rangle\cong C_2^3$, use

$$(X,Y,Z)=(P_0,P_1,P_2),\ (P_0,P_1,P_3),\ (P_2,P_1,P_3).$$

Indeed $P_1P_2=P_2\cup dP_2$ and $P_1P_3=P_3\cup cP_3$. The extra representative belongs to $X\cap Y$ in each chosen triple, so IP immediately gives (3).

For $(X,Y,Z)=(P_0,P_2,P_3)$ the complete finite S_4 coset decomposition is

$$P_2P_3=P_3\cup cP_3\cup bcP_3\cup abcP_3.$$

The first three representatives lie in $P_0\cap P_2$, so their intersections with P_0 are in the required product. The last coset has empty intersection: P_3 fixes $(0,0)$ in the proved action, whereas abc sends it to $(1,0)$. Its P_0-orbit is exactly $\{(0,0),(3,0),(2,0),(1,-1)\}$, by the three explicit generator formulas. For $s\geq 3$ this excludes $(1,0)$. Hence (3) holds for every triple of maximal parabolics.

For completeness these global and local triple properties imply full four-coset Helly. Normalize one coset to P_0. Its intersections with the other three cosets are cosets in P_0. Any two meet by the global triple property, and the local triple property in P_0 gives a common point of all four. Smaller families are already covered. UA.Geometry therefore gives thinness, residual connectivity and flag transitivity, and identifies E_s with the full type-preserving automorphism group, acting regularly on chambers.

Theorem 1. *For every $s\geq 3$, the literal presentation in [1, Theorem 3.1] defines E/sD of order $48s^3$. Its coset system is the universal locally toroidal regular rank-four hypertope specified there, with exact toroidal parameters $(2,0)$ and $(s,0)$ and the source's faithful degree-4s CPR graph.*

Universality means the exact diagram presentation with the stated local relations: Lemma 1 proves a marked isomorphism, not merely the existence of a finite image.

8 The square family and its affine group

We use column coordinates and composition $(A,t)(B,u) = (AB, t + Au)$. The source writes Euclidean transformations on the right. Reversing an involutory word takes its inverse, so its order and its defining power relation are unaffected by this convention. Let $e_1, \dots, e_4$ be the standard Euclidean basis, and put

$$h = \tfrac{1}{2}(e_1 + e_2 + e_3 + e_4), \qquad \Lambda = \mathbb{Z}e_1 + \mathbb{Z}e_2 + \mathbb{Z}e_3 + \mathbb{Z}h.$$

All integer vectors below are coordinates in the ordered basis (e_1, e_2, e_3, h) of Λ.

Take the five source reflections, here specified in Euclidean coordinates:

$$\begin{aligned}
R_0(x) &= (1 - x_1, x_2, x_3, x_4),\\
R_1(x) &= x - \tfrac{1}{2}(x_1 - x_2 - x_3 - x_4)(1, -1, -1, -1),\\
R_2(x) &= (x_1, x_2, x_3, -x_4),\\
R_3(x) &= (x_1, x_2, x_4, x_3),\\
R_4(x) &= (x_1, x_3, x_2, x_4).
\end{aligned}$$

Set $E = \langle R_0, \dots, R_4 \rangle$ and mark it by

$$t_0 = R_1, \quad t_1 = R_2, \quad t_2 = R_1R_3, \quad t_3 = R_0R_4. \tag{S.1}$$

These transformations preserve Λ. They generate E: besides $R_1 = t_0$ and $R_2 = t_1$, one has

$$R_3 = t_0t_2, \qquad R_4 = (t_0t_3)^3, \qquad R_0 = t_3(t_0t_3)^3. \tag{S.2}$$

The last two identities follow from the commuting pair R_0, R_4, the commuting pair R_1, R_4 and $(R_1R_0)^3 = 1$.

For $J \subseteq \{0,1,2,3\}$ write $E_J = \langle t_j : j \in J \rangle$. Let Q_J be its linear image and K_J its translation kernel. The full linear image Q has order 1152. A faithful finite action for its calculation is the action on the following 48 Euclidean roots:

$$\{\pm e_i\} \ \cup \ \{\tfrac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4)\} \ \cup \ \{\pm e_i \pm e_j : i < j\}. \tag{S.3}$$

The roots span the space, so fixing them all forces a linear map to be the identity.

The exact affine computation gives the following complete table. The two nonsingleton translation lattices have bases displayed below it.

J	$\lvert Q_J\rvert$	K_J	J	$\lvert Q_J\rvert$	K_J
$\varnothing$	1	0	$\{3\}$	2	0
$\{0\}$	2	0	$\{0,3\}$	12	0
$\{1\}$	2	0	$\{1,3\}$	4	0
$\{0,1\}$	6	0	$\{0,1,3\}$	48	0
$\{2\}$	2	0	$\{2,3\}$	6	0
$\{0,2\}$	4	0	$\{0,2,3\}$	36	0
$\{1,2\}$	12	0	$\{1,2,3\}$	36	K
$\{0,1,2\}$	48	0	$\{0,1,2,3\}$	1152	$\mathbb{Z}^4$

Here

$$K = \mathbb{Z}u + \mathbb{Z}v, \qquad u = (1,-1,1,0), \quad v = (0,-1,0,1). \tag{S.4}$$

In particular the full translation lattice of E is exactly Λ. The entries are obtained by enumerating each finite Q_J, retaining affine word lifts and computing all normalized Schreier translations, as in AL.1 of the common paper. Integer Hermite reduction computes their entire generated

lattice. It is essential here to use normalized transitions: a list of unnormalized differences alone would not suffice.

The table also exhibits an integral direct complement to K. Writing f_i for the integer coordinate vectors, the four columns u, v, f_2, f_3 form a unimodular matrix. More explicitly, every $x = (x_1, x_2, x_3, x_4) \in \mathbb{Z}^4$ has the unique expression

$$x = x_1 u + x_4 v + (x_2 + x_1 + x_4) f_2 + (x_3 - x_1) f_3. \tag{S.5}$$

Consequently $K \cap s\mathbb{Z}^4 = sK$ for every positive integer s.

9 The presentation and the period

Put $a = t_0, b = t_1, c = t_2, d = t_3$. Let $\mathcal{U}$ be the abstract group on four generators with their involution relations and

$$\begin{gathered}(ab)^3 = (ac)^2 = (ad)^6 = (bc)^6 = (bd)^2 = (cd)^3 = 1, \\ \big(c(da)^2\big)^2 = \big(b(ad)^2 a\big)^2 = \big(a(bc)^2 b\big)^2 = 1.\end{gathered} \tag{S.6}$$

These are the square diagram relations and the three fixed toroidal residue relations. All hold in the affine transformations. The marked map $\mathcal{U} \to Q$ is onto. Its exact Reidemeister–Schreier presentation, followed by Tietze simplification, has four kernel generators and the following six relators:

$$\begin{aligned} &z_4^{-1} z_1^{-1} z_4 z_1, \quad z_2 z_1^{-1} z_2^{-1} z_1, \quad z_4^{-1} z_2 z_4 z_2^{-1}, \\ &z_3 z_1 z_3^{-1} z_1^{-1}, \quad z_3^{-1} z_4^{-1} z_3 z_4, \quad z_3^{-1} z_2 z_3 z_2^{-1}. \end{aligned} \tag{S.7}$$

Thus the actual kernel, not merely its abelianization, is free abelian of rank four. The affine translation kernel has the same rank by the preceding table. AL.2 now identifies $\mathcal{U}$ with E, with the marking preserved.

Define $w = d(cb)^2$. Direct affine multiplication gives

$$w^2 = (I, v). \tag{S.8}$$

The Q-orbit of v consists of 24 vectors and spans $\mathbb{Z}^4$ over the integers. This is checked by its full integral Hermite basis I_4. Conjugating a translation by an affine transformation applies its linear part to the translation vector. Hence the normal closure of w^{2s} in E is exactly $s\Lambda$, and

$$\mathcal{U} / \langle\langle w^{2s} \rangle\rangle \cong E / s\Lambda, \qquad |E / s\Lambda| = 1152 s^4. \tag{S.9}$$

This proves a marked presentation identity for all s at once.

For comparison with the source's original toroid, the five-generator Coxeter presentation $[3, 3, 4, 3]$ has the same affine realization. The separate exact kernel calculation over Q again reduces to four generators with all six pairwise commutator relations and no others. AL.2 identifies that presentation with E. Its period word satisfies

$$\big(R_0 R_1 (R_2 R_3 R_4)^3\big)^2 = (I, f_4).$$

The Q-orbit of f_4 also spans $\mathbb{Z}^4$. Its sth power therefore imposes exactly $s\Lambda$. Together with (S.2), this proves that the groups in Proposition 4.1 are the same marked constructions under the displayed change of generators, not just groups of the same order.

10 The ordered residues

Let P_i be the image of $E_{I\setminus\{i\}}$ in $E/s\Lambda$. For $i = 1, 2, 3$ the linear maps are injective on the entire affine parabolics, so their groups do not collapse in any translation quotient. In the respective ordered paths

$$(t_2, t_3, t_0), \qquad (t_1, t_0, t_3), \qquad (t_0, t_1, t_2),$$

the edge orders are $3, 6$ and the end generators commute. Their additional relations in (S.6) are precisely those of $\{3, 6\}_{(1,1)}, \{3, 6\}_{(2,0)}, \{3, 6\}_{(2,0)}$. The standard triangle presentations with these relations have orders $36, 48, 48$, respectively; exact finite presentation calculations give these orders. The table gives the same orders for the three images, so the marked epimorphisms are isomorphisms. No classification by order alone is being used.

For P_0, put $(A, B, C) = (t_3, t_2, t_1)$ and let

$$\Delta = \langle A, B, C \mid A^2 = B^2 = C^2 = (AB)^3 = (BC)^6 = (AC)^2 = 1\rangle.$$

Its epimorphism onto the actual linear group $Q_{\{1,2,3\}}$ of order 36 has kernel with the exact simplified presentation

$$\langle z_1, z_2 \mid z_2 z_1^{-1} z_2^{-1} z_1 = 1\rangle.$$

The affine image has translation lattice K of rank two. The restriction from this kernel onto K is a surjection of free abelian groups of the same rank, hence an isomorphism. This is the same rank argument as AL.2, now with rank two inside a four-dimensional ambient affine space. Thus $E_{\{1,2,3\}}$ is the marked triangle group Δ itself.

The $Q_{\{1,2,3\}}$-orbit of v has six vectors and spans exactly K. It follows that imposing $(A(BC)^2)^{2s} = 1$ on Δ factors its translation lattice by sK. By (S.5) the ambient quotient induces exactly the same subgroup sK, without any extra residue collapse. This is the standard marked presentation of $\{3, 6\}_{(s,s)}$, and its order is $36s^2$. The ordered rank-three groups are therefore exactly the four claimed in Proposition 4.1.

The source also computes two word orders to distinguish the possible residue periods. Here (S.8) has order s, while $A(BC)^2B$ has linear part of order three and cube (I, u). The primitive vector u has order s modulo $s\Lambda$, so this word has order $3s$. The word itself is generally not a translation: it is its cube that is a translation. This corrects the terminology in the source without changing either asserted order.

11 Intersections and flag transitivity for every period

The fixed linear marking has the intersection property. The certificate checks all 16^2 equalities $Q_J \cap Q_T = Q_{J\cap T}$ as literal sets of matrices. Its four maximal parabolic coset sets have $32, 32, 24, 24$ members. Complete enumeration of pairwise incident cosets gives the following numbers after translating the first coset to the subgroup itself:

types	012	013	023	123	0123
pairwise incident families	18	18	18	18	36.

Every listed family has a common element. Rank-one and rank-two families already satisfy that assertion by nonemptiness and incidence. This proves typed coset Helly for the fixed linear quotient. The normalization loses no family, because right translation acts transitively on the cosets of the first type.

Now fix a positive integer s and set $G = E/s\Lambda$, with the induced map $q : G \to Q$. Its kernel is $N = \Lambda/s\Lambda$. For a proper marked parabolic G_J, the table and the fact that $s\Lambda$ consists of translations give

$$N \cap G_J = \begin{cases} (K + s\Lambda)/s\Lambda, & J = \{1, 2, 3\}, \\ 1, & \text{every other proper } J. \end{cases} \tag{S.10}$$

For the full index set the intersection is N. This entire kernel profile has the intersection property: two different proper supports cannot both have a nontrivial kernel intersection. UA.K therefore proves the intersection property in G.

Equation (S.5) yields the internal direct product

$$N \cong (K/sK) \ \oplus \ \big((\mathbb{Z}f_2 + \mathbb{Z}f_3)/s(\mathbb{Z}f_2 + \mathbb{Z}f_3)\big).$$

Each maximal parabolic intersection in (S.10) is a product of some of these two factors. The linear quotient has IP and typed coset Helly, so UA.SK.Helly transfers typed coset Helly to G. Its four markers have order exactly two, as their nontrivial linear images do. UA.Geometry now proves that the coset incidence system is thin, residually connected and flag-transitive, and that its full type-preserving automorphism group is G.

Finally (S.6) and (S.9) give exactly the diagram relations and the four ordered residue relations. There is no extra global relation. This proves universality, completing Theorem 4.3 for its full stated range $s \geq 2$. The argument in fact also verifies these group and coset assertions at $s = 1$; no additional claim about the terminology for the source's rank-five toroid at that parameter is needed.

12 The literal presentations and a small countermodel

Here the order of the markers is $a = \rho_0, b = \rho_1, c = \rho_2, d = \rho_3$. Let B denote the following thirteen words, all imposed as relators:

$$a^2, b^2, c^2, d^2, (ab)^3, (ac)^6, (ad)^2, (bc)^2, (bd)^6, (cd)^3,$$
$$(d(ca)^2c)^2, \quad (a(bd)^2)^2, \quad (b(ac)^2a)^2.$$

Put $w = c(db)^2d$ and $v = c(db)^2$. For $t \in \{1, 2\}$ define

$$P_t = \langle a, b, c, d \mid B, w^{3t} \rangle, \qquad V_t = \langle a, b, c, d \mid B, v^{2t} \rangle. \tag{F.1}$$

Thus P_1, P_2 are the original presentations. The printed replacement on p. 10 has an extra closing parenthesis, but removes the final d and uses exponent two; p. 11 unambiguously uses v^4. The assertions that these replacements imply $(bd)^2 = 1$ and produce the stated 240-chamber polytope are false for these literal words.

Here is one countermodel for both, in the symmetric group on eleven points; multiplication is functional composition, right factor first:

$$a = (1\,3)(2\,4)(5\,6)(7\,9), \qquad b = (1\,2)(3\,5)(4\,6)(7\,8),$$
$$c = (3\,6)(4\,5)(9\,10), \qquad d = (1\,4)(2\,3)(10\,11).$$

Direct substitution makes every word of B, v^2, and v^4 the identity, while bd has order six. In particular $(bd)^2$ sends 1 to 4. Closing these four permutations under multiplication gives 720 distinct elements. Hence neither V_t can have order 240: each has a quotient of order 720. The complete integer-permutation check is `finite/countermodel.py`. Six declarations in `finite/formal/SectionFive.lean` prove the thirteen shared relations, both replacement relations, the failure of $(bd)^2 = 1$, and both resulting universal nonconsequences. The order 720 computation is not included in the Lean scope.

13 Exact corrected groups and ordered residues

Within P_t, for an omitted index i let G_i be the subgroup generated by the other three markers. We write $T(s, 0)$ for the marked triangular torus group with presentation

$$\langle x, y, z \mid x^2, y^2, z^2, (xy)^3, (yz)^6, (xz)^2, (x(yz)^2y)^s \rangle,$$

and $T(t,t)$ for the same first six relators followed by $(x(yz)^2)^{2t}$. These are the usual triangle-reflection torus conventions in [1]; their orders are $12s^2$ and $36t^2$.

Theorem 2. *For $t = 1, 2$, the presentations P_t and V_t are isomorphic by the map fixing all four named generators, and $|P_t| = 720t^6$. Their ordered maximal parabolics are as follows.*

Omitted index	*Ordered markers (x, y, z)*	*Marked group*	*Order*
0	(c, d, b)	$T(t,t)$	$36t^2$
1	(d, c, a)	$T(2,0)$	*48*
2	(a, b, d)	$T(1,1)$	*36*
3	(b, a, c)	$T(2,0)$	*48*

The marked map $T(3t, 0) \longrightarrow G_0$ is an epimorphism with kernel of order three. The associated torus-map covering has degree three.

Proof. The complete required program `finite/exact_groups.g` constructs each abstract group in (F.1) and calls GAP's exact fp-group isomorphism routine, not just a homomorphism onto an assumed finite image. It obtains orders 720 and 46080. In the original presented group it verifies $v^{2t} = 1$; in the modified presented group it verifies $w^{3t} = 1$. Thus the two identity-on-generators homomorphisms exist and are inverse. Exact GAP fp-group and coset computations are declared computational premises of this proof.

For each row of the table, the program constructs the displayed three-generator abstract torus presentation, checks its order, and constructs a homomorphism to the actual parabolic with exactly the listed generator images. This map is onto and both groups have the listed order, so it is a marked isomorphism, not identification from orders alone. The analogous map from $T(3t, 0)$ is onto G_0; its domain order $108t^2$ gives kernel size three. These are the standard triangular translation lattices, with no local rank-two collapse: each relevant marked dihedral subgroup has its prescribed order. Hence the torus map is a covering of degree three.

For a second, elementary finite check, `finite/model_data.json` records faithful images on 13 and 30 points. `finite/finite_models.py` closes these literal permutations, substitutes every original and modified relator, and verifies all parabolics and all tests below. Its closure gives the finite-image lower bound only; the abstract upper bound still comes from the separate exact fp-group computation above. □

14 Intersection, flags and universality

Lemma 3. *For three subgroups H, K, L of a group, all pairwise intersecting triples of their right cosets have a common point if and only if*

$$H \cap KL = (H \cap K)(H \cap L). \tag{F.2}$$

If four subgroup types satisfy this for every triple, and the three intersection subgroups inside any one type also satisfy it, every pairwise intersecting four-coset family has a common point.

Proof. Translate an intersecting pair of cosets to H, K. Choose $h \in H \cap Lx$ and $k \in K \cap Lx$. Then $hk^{-1} \in L$, so $h \in H \cap LK$. Taking inverses in (F.2) gives $h = l_0k_0$ with $l_0 \in H \cap L$ and $k_0 \in H \cap K$. Now $k_0 = l_0^{-1}h \in Lx$ is a common point. Conversely the common-point property applied to these normalized triples gives the factorization. For four cosets, intersect each of the other three with a fixed fourth. Global triple intersection makes these three cosets inside the fourth pairwise intersecting. The local three-coset property supplies their common point. □

For each of the two actual finite groups, both required checks enumerate all 16 generated subgroups G_J, including $G_\varnothing = 1$, and test all 256 equations

$$G_J \cap G_K = G_{J \cap K}. \tag{F.3}$$

They also test (F.2) for all four triples of maximal parabolics, and for the three subgroups $G_i \cap G_j$ inside each G_i (four more tests). All equalities hold as full sets, not merely as cardinalities. Lemma 3 therefore proves the coset geometry flag-transitive in every rank up to four.

The markers are nontrivial involutions. Equation (F.3) and the full typed coset Helly property therefore satisfy UA.Geometry of the common paper. The coset system is thin, residually connected and flag-transitive, and the displayed group is its full type-preserving automorphism group.

The modified presentation V_t comprises precisely the circuit relations and the four displayed ordered rank-three period relations. By the marked isomorphism, P_t is therefore also the universal marked group with those local presentations. In particular an alternative compatible marked regular geometry is a quotient; the local maps are isomorphisms when its parabolics have those exact presentations. Neither literal replacement changes this group. The distinction between ordered local constraints and an unordered list of toroidal maps is retained throughout; we do not claim to classify every permutation of the local data.

15 The relation with the full cubic toroid

Let Q be the signed permutation group on three coordinates and $D = \{v \in \mathbb{Z}^3 : v_1 + v_2 + v_3 \equiv 0 \pmod 2\}$. In the notation of the reviewed cubic-family proof, the hypertope group is $E_s = E/sD$, where

$$E = \{(q, v) \in Q \ltimes \mathbb{Z}^3 : \textstyle\sum v_i \equiv \epsilon(q) \pmod 2\}, \qquad |E_s| = 48s^3.$$

The introductory sentence on page5 of [1] identifies this group with the full group of the cubic toroid $\{4, 3, 4\}_{(s,s,0)}$. The earlier account [2, pp. 14–15] correctly describes an index-two subgroup. Here is the exact change of generators and coordinates.

The standard Euclidean realization of $[4, 3, 4]$ is generated by

$$R_0(x) = (1 - x_1, x_2, x_3), \qquad R_1(x) = (x_2, x_1, x_3),$$
$$R_2(x) = (x_1, x_3, x_2), \qquad R_3(x) = (x_1, x_2, -x_3).$$

Its affine image is $W = Q \ltimes \mathbb{Z}^3$: the last three generators give Q, and multiplication of R_0 by the corresponding linear reflection gives translation by e_1. Conjugating gives all coordinate translations. The faithful Euclidean Coxeter realization is a retained classical fact. Direct multiplication gives

$$(R_0R_1R_2R_3R_2)^2 = (I, e_1 + e_2).$$

Its Q-orbit generates D. Thus the additional relation with exponent $2s$ used in [2] imposes precisely sD, and the full toroid group is W/sD, of order $48[\mathbb{Z}^3 : sD] = 96s^3$.

Put $p = (1/2, 1/2, 1/2)$ and $T(x) = \mathrm{diag}(1, -1, -1)(x - p)$. Conjugating the four displayed ambient words by T gives

$$T(R_0R_1R_0)T^{-1}(x) = (x_2, x_1, x_3),$$
$$TR_2T^{-1}(x) = (x_1, x_3, x_2),$$
$$T(R_1R_3)T^{-1}(x) = (-x_2, -x_1, 1 - x_3),$$
$$TR_1T^{-1}(x) = (-x_2, -x_1, x_3).$$

These are exactly the four generators of E in source-label order. The linear part of T preserves D, so this conjugacy descends to the period quotients. Consequently the displayed subgroup of W/sD is isomorphic to E/sD and has index two. In [2] the Petrie generators are $\alpha_0 = R_0$, $\alpha_1 = R_1$, $\alpha_2 = R_2$ and $\alpha_3 = R_1R_3$; its four subgroup generators are the same set, ordered here as $(\alpha_1^{\alpha_0}, \alpha_2, \alpha_3, \alpha_1)$ to match [1].

The family still arises from the stated toroid. The correction is that its group is the displayed index-two subgroup, rather than the full toroid group. This changes no principal result of Section3 of [1]. It does not claim a new family or a resolution of an open question.

Whole dependencies and formal limits. Certificate paths in the transferred sections refer to the entire corresponding files under `inherited/`; their exact source bytes are preserved. The new finite-row engine is separately listed in the application. GAP's exact finite-presentation, coset and permutation algorithms and the bounded ACE enumeration are declared software premises. The inherited six Lean declarations concern only the finite countermodel and nonconsequences. They are not a formal proof of the general affine or C-plus geometry theorems. No numerical or source-credit assertion is made by this relocation.

References

[1] M. E. Fernandes, D. Leemans, C. A. Piedade and A. Ivić Weiss, *Two families of locally toroidal regular 4-hypertopes arising from toroids*, retained twelve-page author edition.

[2] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, retained 21-page author edition.

[3] *Supported kernels, translation lattices and finite marked quotients*, complete accepted common manuscript, AL.1–AL.2, AA.ParabolicPeriods, UA.K, UA.SK.Helly and UA.Geometry.